



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

DIFFERENTIAL AND INTEGRAL CALCULUS

1M — 104018

WINTER 2021

Exam B

February 21, 2022

Student ID Number:

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Instructions

1. **Duration: 180 minutes**
2. This exam has **5 questions**, for a total of **100** points.
3. Write your answers clearly, completely and concisely in the space provided.
4. **Quote** all results that you use. **No credit will be given to answers which are not justified.**
5. Use of calculators, personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed.
6. During the exam, a dictionary will be available upon your request to the proctors.

Question 1 (10+10 points)

- (a) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and define $h(x) = f(2x + g(x))$, for all $x \in \mathbb{R}$. Knowing that $f(0) = 1$, $f(1) = 3$, $f'(1) = 2$, $g(0) = 1$, $g(1) = 2$ and $g'(0) = 3$, determine the equation of the tangent line to the graph of h at the point $(0, h(0))$.
- (b) Compute: $\lim_{x \rightarrow 0^+} x^{e^x - 1}$.

Answer

Question 2 (8+8+8 points) Compute the following integrals:

(a) $\int x^3 \sqrt{1-x^2} \, dx;$

(b) $\int_0^M \frac{1}{(x+1)(x^2+1)} \, dx$, where $M > 0$;

(c) $\int_0^{+\infty} x e^{-x^2} \, dx.$

Answer

Question 3 (20 points) Find the area of the region (in the first quadrant) bounded by the curves $y = \sqrt{x} + 1$, $y = 2x^2$ and $y = 1 - x$.

Answer

Question 4 (7+7+7 points)

(a) Compute the sum of the series $\sum_{n=1}^{\infty} \left(\frac{2^{2n-1} + 7 \cdot 3^{n+1}}{5^n} \right)$.

(b) Study the convergence of the series $\sum_{n=1}^{\infty} \frac{n^{20} 3^n}{(2n)!}$.

(c) Determine the radius of convergence and the interval of convergence of the power series $\sum_{n=1}^{\infty} \frac{(-2)^{n-1}}{n} x^n$.

Answer

Question 5 (9+6 points) Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous and differentiable on $(0, 1)$, with $f'(x) \leq M$, for all $x \in (0, 1)$, where $M = f(1) - f(0)$.

(a) Show that $f(x) \leq f(0) + Mx$, for all $x \in [0, 1]$.

[HINT: Show that the function $g(x) = f(x) - Mx$ is decreasing.]

(b) Conclude that $f(x) = f(0) + Mx$, for all $x \in [0, 1]$.

Answer

